

Expectation-Maximization Driven Contrastive Disentanglement for Generalized Category Discovery

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Abstract

Generalized Category Discovery (GCD) is a critical task in open-world computing scenarios, aiming to automatically classify partially labeled data by recognizing both known and novel categories. However, existing GCD methods usually suffer from inherent bias toward known categories due to the exclusive pre-training on them and the absence of labeled data of novel categories. This bias can lead to significant misclassification and clustering errors for novel categories. Although recent approaches leverage pseudo-label training and contrastive learning to address this, they still lack explicit supervision to disentangle novel and known categories, resulting in performance bottlenecks. To address these limitations, we propose an Expectation-Maximization-driven Contrastive Disentanglement (EMCD) framework designed to explicitly disentangle novel and known categories. We particularly formulate the identification of novel categories as a latent variable estimation problem. Specifically, it incorporates an EM-disentangling regularization to softly identify novel category samples and a consistency regularization to enhance generalization. In addition, we leverage dual contrastive constraints, including a cluster-sample contrastive constraint and a sample-sample contrastive constraint, to pull close samples of novel categories while pushing apart ambiguous samples near decision boundaries. Empirical results on 3 commonly used datasets demonstrate that our model is effective and outperforms previous state-of-the-art methods. Our code is available at <https://github.com/YWY-only/EMCD>.

CCS Concepts

• **Computing methodologies** → **Natural language processing**;
Semi-supervised learning settings.

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Keywords

Generalized Category Discovery, Expectation-Maximization, Contrastive Learning, Category Disentanglement

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1 Introduction

Generalized Category Discovery (GCD) serves as a central task in open-world computing scenarios [16, 26], with the aim of automating the classification of partially labeled data. Specifically, the task aims not only to recognize known categories, but also to discover completely novel categories, using a mix of small labeled data and large unlabeled data [4, 15]. This capability mirrors the dynamic evolution of real-world data, where emerging categories require adaptive continuous learning frameworks.

Most existing GCD methods [1, 2, 16, 26] follow a two-stage pipeline: First, they pre-train a model (e.g., BERT [9]) using both supervised learning on labeled data and self-supervised learning on unlabeled data. Then, clustering algorithms (e.g., K-Means) are applied to group samples into known and novel categories. However, as illustrated in Figure 1, this paradigm suffers from three types of critical error, particularly within novel categories. A fundamental challenge stems from the model's inherent bias toward *known categories*, due to the exclusive pre-training on them, coupled with the absence of supervision for novel categories. Consequently, samples from novel categories are frequently misclassified as known categories or incorrectly clustered with other novel categories, which severely impedes the effective discovery of novel categories.

To address model bias and acquire knowledge about novel categories from unlabeled data, previous studies explore pseudo-label training [16, 25, 28] and contrastive learning methods [13, 26, 29]. Among them, contrastive learning-based methods become a dominant GCD paradigm, which learn discriminative representations through minimizing intra-class variance and maximizing inter-class

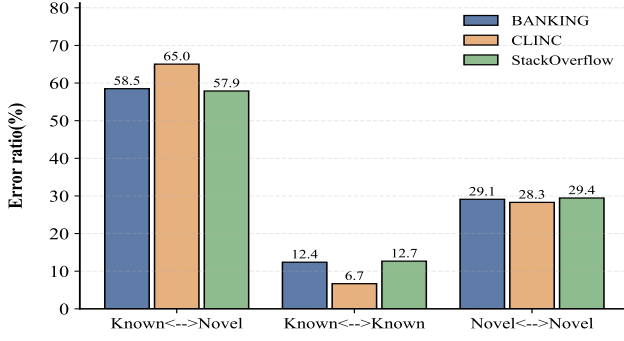


Figure 1: Error ratios (%) of a pre-trained classifier on three datasets. Known↔Novel denotes samples from known categories and novel categories are misclassified. Known↔Known denotes internal misclassification of samples within known categories. Novel↔Novel denotes internal misclassification of samples within novel categories.

separation. To further distinguish different categories, An et al. [1] implement neighborhood contrastive learning with LLMs, focusing on unlabeled samples near decision boundaries. Moreover, An et al. [2] employ category prototypes to measure category similarities and facilitate knowledge transfer. Nevertheless, these methods lack an explicit distinction supervision between novel and known categories, which can still suffer from the bias towards known categories, leading to tangling and confusion between them.

Unlike the previous methods, we propose an explicit learning framework to disentangle novel and known categories. Specifically, we formulate the identification of novel categories as a latent variable estimation problem. We initially separate novel and known categories using an entropy-based criterion, where samples from novel categories typically exhibit higher predictive entropy under the classifier trained solely on known categories. This entropy measure is then used as a supervision signal to guide the learning of the latent variable. The above learning process is achieved by an Expectation-Maximization (EM) algorithm, which iteratively refines the latent variable estimation by maximizing the expected likelihood of novel category identification. In this way, the model explicitly learns to discriminate between novel and known categories, thereby alleviating the confusion between them.

Following the above idea, we propose an EM-driven Contrastive Disentanglement framework for GCD, namely EMCD. Based on the pre-trained base model, we propose a latent category disentanglement module to distinguish novel and known categories. In particular, it contains an EM-disentangling regularization to softly identify samples of novel categories, and an imposed consistency regularization to improve the generalizability of the EM disentanglement. On the other hand, we also propose an explicit category disentanglement module to further improve representation clustering. Specifically, we devise a cluster-sample contrastive constraint to pull close the samples of novel categories, and a sample-sample contrastive constraint is also imposed on the samples near category decision boundaries, pulling apart different categories. The main contributions of this work can be summarized as follows:

- We propose an EM-driven contrastive disentanglement GCD framework, EMCD, to disentangle novel and known categories, which alleviates the critical bias towards known categories.
- To further disentangle different categories in clustering, we devise a cluster-sample contrastive constraint and a sample-sample contrastive constraint, pulling close the samples of novel categories and pulling apart between different categories.
- Empirical studies indicate that our model outperforms previous state-of-the-art methods on 3 commonly-used benchmarks, and our model version without LLMs can even achieve competitive results with existing LLM-based methods.

2 Related Work

In this section, we introduce the background of the generalized category discovery and the expectation-maximization algorithm.

2.1 Generalized Category Discovery

Generalized Category Discovery (GCD) was proposed and widely studied in both NLP [3, 26] and computer vision [21, 22]. GCD is a recently emerged task that addresses a realistic scenario where only limited labeled data are available and novel categories may emerge, with the goal of automatically clustering all unlabeled data from both known and novel categories. Pioneering works [1, 15, 21, 22] mainly employ a two-stage scheme involving base model pretraining and iterative optimization followed by clustering methods such as K-Means. To address model bias and enhance the acquisition of knowledge about novel categories from unlabeled data, there exist two major practices:

- *Pseudo-labeling based* methods [16, 25, 28] pseudo labels for unlabeled data based on pretrained embeddings. For example, [18] proposed transferring knowledge from known to novel categories through label adjustment.
- *Contrastive learning based* methods [1, 13, 14, 26, 27, 29] learn text sample embeddings by different clustering strategies. For example, Zhang et al. [26] addresses the clustering inconsistency problem by using an alignment strategy to learn text embeddings.

Although these methods are all aimed at alleviating the bias of novel categories towards known categories, they implicitly enhance the distinguishability between known and unknown categories. Due to persistent bias, suboptimal results are achieved in novel categories. However, our method converts this problem into an explicit solvable one through the hope maximization of latent variables.

2.2 Expectation-Maximization Algorithm

The Expectation-Maximization (EM) algorithm is an iterative approach for parameter estimation in statistical models that incorporate latent variables. In cases where the true values of these latent variables remain unknown, maximum likelihood estimation (MLE) cannot be used to identify the model parameters. Instead, the EM algorithm uses Jensen’s inequality to maximize the lower bound of

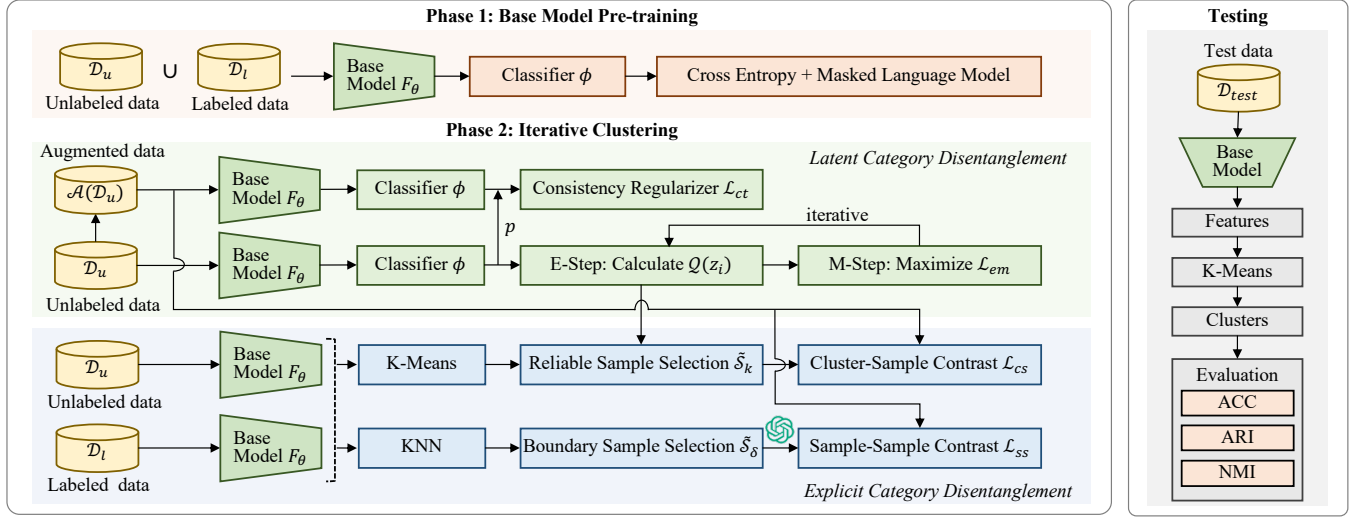


Figure 2: Overview of the framework, EMCD. It includes two phases for training. We propose a latent category disentanglement module to explicitly distinguish novel categories, and an explicit category disentanglement module to improve clustering.

the log-likelihood function:

$$\begin{aligned}
 \log p(y|\theta) &= \log \sum_z p(y, z|\theta) \\
 &= \log \sum_z p(z|y, \theta) \frac{p(y, z|\theta)}{p(z|y, \theta)} \\
 &\geq \mathbb{E}_{z \sim p(z|y, \theta)} \left[\log \frac{p(z|y, \theta) p(y, z|\theta)}{p(z|y, \theta)} \right] \\
 &= \sum_z p(z|y, \theta) \log \frac{p(y, z|\theta)}{p(z|y, \theta)} := \mathcal{L}(\theta),
 \end{aligned} \tag{1}$$

where z , y , and θ denote the latent variable, observation, and model parameter, respectively. The expectation of the conditional probability distribution $p(z|y, \theta)$ for the latent variable z is called Q function. Then, the loss $\mathcal{L}$ can be written as

$$\mathcal{L}(\theta) = \sum_z Q(z_i) \log \frac{p(y, z|\theta)}{Q(z_i)}. \tag{2}$$

The algorithm alternates between the two steps:

- **Expectation step (E-step):** Sample latent variables from the current estimation of $Q(z_i)$, and compute the expected log-likelihood lower bound $\mathcal{L}(\theta)$.
- **Maximization step (M-step):** Optimize parameters θ by maximizing Eq. (2) for a better estimation of $Q(z_i)$.

This iterative procedure allows the EM algorithm to converge to a local maximum of the log-likelihood of the observed data, which makes it powerful for estimation problems involving latent variables, such as Gaussian mixture clustering [17], and dynamical system identification [8, 19]. The probabilistic framework of Zhou et al. [31] applies the EM algorithm with category membership as a latent variable in a joint optimization scheme. Unlike other approaches, our proposed framework explicitly decouples the processing of known and novel categories using the EM algorithm, allowing for a more targeted and effective estimation.

3 Task Formulation

Formally, we have an open-world dataset $\mathcal{D}$, including a small labeled set $\mathcal{D}_l$ containing only known categories, and a large unlabeled set $\mathcal{D}_u$ containing both known and novel categories. Let $\mathcal{D}_l = \{(x^l, y^l)\}$ be the set of pre-labeled text samples, where each x^l and $y^l \in \mathcal{C}$ denote a text sample and its label, respectively. Let $\mathcal{D}_u = \{x^u\}$ be the set of unlabeled text samples, where x^u denotes an unlabeled text sample. Let $\mathcal{C} = \mathcal{C}_K \cup \mathcal{C}_N$ be the total potential categories, where $\mathcal{C}_K$ and $\mathcal{C}_N$ are the set of known and novel categories, respectively. The number of total categories $|\mathcal{C}|$ is considered a known prior [1, 22]. The goal of GCD is to accurately cluster unlabeled data in $\mathcal{D}_u$ to decide the corresponding categories, using the limited supervision information from $\mathcal{D}_l$.

4 Method

Our framework is presented in Figure 2. In particular, we propose a latent category disentanglement module to distinguish novel and known categories, and an explicit category disentanglement to distinguish different clustering categories.

4.1 Base Model

Following previous studies [1, 4, 15], we use the lightweight language model BERT [9] as the base model F_θ to extract feature representation f_i for the input sentence x_i as

$$f_i = F_\theta(x_i). \tag{3}$$

For GCD, general methods typically conduct two procedures:

(1) **Base model pre-training:** This procedure obtains better representation for clustering in GCD. Specifically, we pre-train F_θ on both labeled and unlabeled data in a multitask manner [29], which uses cross-entropy (CE) to learn labeled data and mask language modeling (MLM) [9] to learn both labeled and unlabeled data. Through pretraining, F_θ acquires knowledge of known categories

and general knowledge about the data, providing informative initial sample representations.

(2) **Iterative clustering:** This procedure obtains categories (both known and novel) for unlabeled data by clustering. Specifically, we perform K-Means on the representations of all unlabeled data, and obtain clusters with centers $\{\mu_k\}_{k=1}^{|C|}$. Here, each cluster can be seen as a *known* or *novel* category. For a sample $x_i \in \mathcal{D}_u$, we locate its corresponding cluster μ_k , which is treated as the category of its pseudo-label $\hat{y}_i$. We then impose further constraints on the representations according to current clusters, and optimize them for better clustering iteratively.

4.2 Latent Category Disentanglement

Since novel categories cannot be previously known for learning, samples of known and novel categories are usually entangled and difficult to distinguish, as shown in Figure 1 (Known $\leftrightarrow$ Novel). To mitigate entanglement between known and novel categories, we frame this identification with a *latent category disentanglement* module that incorporates a binary latent variable.

4.2.1 EM-disentangling Regularization. Formally, for each sample x_i , we propose to assign a latent indicator z_i , such that: $z_i = 1$, if x_i belongs to known categories as $y_i \in C_K$, and $z_i = 0$, if x_i belongs to novel categories as $y_i \in C_N$. Thereafter, we define a conditional distribution $Q(z_i) := Q(z_i|f_i; \theta)$ for z_i , which satisfies:

$$\sum_{z_i \in \{0,1\}} Q(z_i|f_i; \theta) = 1, 0 \leq Q(z_i) \leq 1, \quad (4)$$

where θ denotes the parameters of the base model F_θ , and f_i is the sample representation of x_i . Here, $Q(z_i = 0) + Q(z_i = 1) = 1$. In practice, the value of $Q(z_i)$ can be seen as a probability of indicating whether the sample x_i belongs to known categories. However, this variable cannot be directly measured, since samples of novel categories are not previously given. Therefore, we propose to leverage the EM algorithm to iteratively measure this variable, which alternates between the two steps:

- **E-Step:** In this step, we would like to estimate $Q(z_i)$ for the M-step. Inspired by findings [18], entropy H is a simple yet effective tool to measure $Q(z_i)$. Specifically, samples from known categories usually exhibit low entropy, considering that the biased model is relatively confident in these categories, while samples from novel categories exhibit high entropy. This nature can be exploited to distinguish novel and known categories, and we use it to approximately estimate $Q(z_i)$ for each sample x_i as

$$\begin{aligned} \mathcal{H}(f_i) &= - \sum_{c_k \in C_K} \phi(f_i)_k \log \phi(f_i)_k, \\ Q(z_i) &:= \text{Norm}(\mathcal{H}(f_i)), \end{aligned} \quad (5)$$

where $\text{Norm}(\cdot)$ denotes a normalization to ensure $0 \leq Q(z_i) \leq 1$, and we use min-max normalization, i.e., $\mathcal{H}(f_i)/\log |C_K|$ if $z_i = 0$, and $1 - \mathcal{H}(f_i)/\log |C_K|$, otherwise. Here, ϕ is the pre-trained classifier that maps the representation f_i into known categories C_K . Note that the parameter θ is fixed in estimating $Q(z_i)$.

- **M-Step:** In this step, we calculate $\mathcal{L}(\theta)$ based on $Q(z_i)$, which is further used to optimize the parameterized base model F_θ . With the estimated $Q(z_i)$ in the E-step, we can derive the expected

log-likelihood lower bound $\mathcal{L}(\theta)$ in Eq. (2) to maximize as:

$$\begin{aligned} \mathcal{L}(\theta) &= \sum_{x_i \in \mathcal{D}_u} \sum_{z_i \in \{0,1\}} Q(z_i) \log p(z_i|f_i; \theta) - Q(z_i) \log Q(z_i) \\ &\propto \sum_{x_i \in \mathcal{D}_u} \sum_{z_i \in \{0,1\}} Q(z_i) \log p(z_i|f_i; \theta) := \mathcal{L}_{em}(\theta), \end{aligned} \quad (6)$$

where $Q(z_i) \log Q(z_i)$ is omitted as it is a constant that does not affect optimization. Here, $p(z_i|f_i; \theta)$ is achieved by the classifier $\phi(f_i)$, and $\theta := \arg \max_\theta \mathcal{L}(\theta)$ is optimized and used to update $Q(z_i)$ in the E-step. Therefore, during training, the EM algorithm starts with the pre-trained base model, and then iterates between the E-step and M-step until convergence.

4.2.2 Consistency Regularization. To enhance generalizability, we further apply consistency regularization to the base model with the classifier of known categories. In general, we expect the base model to produce stable and consistent sample representations, reducing the variance of latent variables under minor perturbations. Following the well-known consistency assumption [5], for any unlabeled sample $x_j \in \mathcal{D}_u$, the model is expected to make a consistent prediction on its similar samples. Hence, we impose a data augmentation $\mathcal{A}(\cdot)$ on x_j to derive its similar samples, and enforce the consistent prediction between them. The process is as follows:

$$\mathcal{L}_{ct} = \frac{1}{|\mathcal{D}_u| |C_K|} \sum_{x_j \in \mathcal{D}_u} \sum_{c_k \in C_K} \mathcal{H}(p(c_k|\mathcal{A}(x_j)), p(c_k|x_j)), \quad (7)$$

where we use the promising Random Token Replace [7] to achieve $\mathcal{A}(\cdot)$. In addition, $\mathcal{H}(p_1, p_2)$ is a consistency measurement between p_1 and p_2 , and we use the Kullback-Leibler (KL) divergence [20] for the achievement. In this way, the consistency regularization prevents the model from over-relying on spurious correlations in the training data, which increases the smoothness and robustness of the representations, and can be helpful in distinguishing samples of different categories.

4.3 Explicit Category Disentanglement

Since novel categories are not previously known, the model relies on clustering of sample representations to assign categories. However, known categories only have limited labeled data, and novel categories do not have labeled data. The insufficient supervision makes the learned representation sub-optimal, where the samples can be entangled between different clusters. This would make it difficult to cluster the unlabeled samples accurately, leading to severe errors shown in Figure 1 (Novel $\leftrightarrow$ Novel and Novel $\leftrightarrow$ Known).

To disentangle samples between different clusters, we propose the *explicit category disentanglement* module. In general, we impose the low-density separation principle [7] on the representations of unlabeled samples. Specifically, it includes a cluster-sample contrastive constraint to pull closer unlabeled samples within the same cluster, and a sample-sample contrastive constraint to push apart different clusters. These two constraints help each other and together enhance the separated representations for clustering.

4.3.1 Cluster-Sample Contrastive Constraint. Within the representation clusters obtained by K-means, we expect to *pull close* between samples and the cluster center. Given the cluster centers $\{\mu_k\}_{k=1}^K$, for each sample x_i with its representation f_i , we first measure the

Euclidean distance to each cluster center as:

$$d(x_i, \mu_k) = \|f_i - \mu_k\|_2. \quad (8)$$

Then, for the sample set $\mathcal{S}_k$ of the cluster μ_k , we would like to select reliable samples of each cluster according to the distance as:

$$\tilde{\mathcal{S}}_k = \text{TopK}(d(x_i, \mu_k), m_k), x_i \in \mathcal{S}_k, \quad (9)$$

where $m_k = \lfloor \gamma \times |\mathcal{S}_k| \rfloor$ is the number of samples in cluster μ_k , and γ is the proportion used for all clusters. This approach ensures that the same proportion of samples is selected from each cluster, which serves as reliable samples with high certainty. Here, considering that novel categories usually have low-quality clusters, we particularly focus on these novel clusters as they are more error-prone, and use the reliable samples for supervision. The designed constraint is:

$$\mathcal{L}_{cs} = -\frac{1}{|\mathcal{B}_u|} \sum_{x_i \in \mathcal{B}_u} \mathbb{I}(x_i) \log \frac{\exp(\text{sim}(F_\theta(\mathcal{A}(x_i)) \cdot \mu_i) / \tau_1)}{\sum_{j \in \mathcal{K}} \exp(\text{sim}(F_\theta(\mathcal{A}(x_i)) \cdot \mu_j) / \tau_1)}, \quad (10)$$

$$\mathbb{I}(x_i) = \{x_i | x_i \in \tilde{\mathcal{S}}_k \wedge [Q(z_i = 0) > Q(z_i = 1)]\}, \quad (11)$$

where $\mathbb{I}(x_i)$ are the selected reliable samples that potentially belong to novel categories. τ_1 is a temperature factor. $\mathcal{B}_u$ is the batched subset of the unlabeled data $\mathcal{D}_u$. In this way, samples of potential novel categories (w.r.t., clusters) are selected and optimized to be closer to their corresponding cluster center, leading to more compact representational clusters.

4.3.2 Sample-Sample Contrastive Constraint. Beyond pulling close samples within clusters, we also expect to *push apart* between different clusters. However, directly pushing apart cluster centers can be trivial, as they are naturally separated. Hence, we focus on the ambiguous samples near the decision boundaries between clusters, which are usually hard samples and error-prone. To this end, we first identify the ambiguous samples, and then incorporate them with a sample-sample contrastive constraint.

To identify ambiguous samples from unlabeled samples, we take advantage of local inconsistent sampling (LIS) [1, 30]. Specifically, we select samples that have neighboring samples with different categories. For an unlabeled sample $x_i \in \mathcal{B}_u$, we first find its N -nearest neighbors $\mathcal{N}(x_i)$ from the batched labeled and unlabeled samples $\mathcal{B}$ with Euclidean distance as:

$$\mathcal{N}(x_i) = \text{KNN}(x_i, \mathcal{B}, N), \mathcal{B} = \mathcal{B}_u \cup \mathcal{B}_l, \quad (12)$$

where $|\mathcal{N}(x_i)| = N$. Then, we measure the uncertainty of x_i . Specifically, we treat its belonged categorical cluster μ_k as its pseudo-label $\hat{y}_i$. Then, we define a local inconsistency degree $\delta(\cdot)$ to reflect its uncertainty, which counts the number of neighbors whose pseudo-labels are different from $\hat{y}_i$. It can be formulated as:

$$\delta(x_i) = \sum_{j=1}^N \mathbb{I}(\hat{y}_i \neq \hat{y}_j), x_j \in \mathcal{N}(x_i). \quad (13)$$

The higher δ_i is, the higher the uncertainty of $\hat{y}_i$ tends to be. We then select a number of samples that have high uncertainty, which are treated as the ambiguous samples near decision boundaries:

$$\tilde{\mathcal{S}}_\delta = \text{TopK}(\delta(x_i), m_\delta), x_i \in \mathcal{B}_u, \quad (14)$$

where m_δ is the number of samples selected.

Afterward, we elaborate to confirm the categories of the ambiguous sample in $\tilde{\mathcal{S}}_\delta$, and impose constraints on them to move

away from the decision boundaries. To this end, our main idea is to carefully check its neighbors, and select a reliable sample as the reference sample for contrastive learning. Specifically, we propose three strategies for reliable selection:

(1) **Base model-based selection:** This strategy uses the representation similarity from the base model to decide the categories. Specifically, it measures averaged categorical similarity score of $x_i \in \tilde{\mathcal{S}}_\delta$ to pseudo-labeled samples $x_j \in \mathcal{N}(x_i)$, as:

$$c^* = \arg \max_{c \in \mathcal{C}} \text{score}(x_i, c), \quad \text{score}(x_i, c) = \frac{1}{|\mathcal{S}_{i,c}|} \sum_{x_j \in \mathcal{S}_{i,c}} \text{sim}(f_i, f_j), \quad (15)$$

where $\mathcal{S}_{i,c} = \{x_j | x_j \in \mathcal{N}(x_i) \wedge y_i = c\}$ is the neighboring sample set of x_i with pseudo-labeled category c . Hence, it considers the entire local similarity to decide on a reliable category c^* . We then select the most similar sample with pseudo-label c^* as the reference sample, i.e., $x^* = \arg \max_{x_j \in \mathcal{S}_{i,c^*}} \text{sim}(f_i, f_j)$.

(2) **LLM embedding-based selection:** This strategy leverages powerful LLM embedding models [24] for assistance. Given the sample $x_i \in \tilde{\mathcal{S}}_\delta$ and its neighbors $\mathcal{N}(x_i)$, using the base model, we select the neighbors with the highest similarity of each category, denoted $\mathcal{O}$. Then, we generate sample embeddings with the LLM, where the prompt of x_i is "Retrieve semantically similar text. Query: $[x_i]$ ", and that of other sample $x_j \in \mathcal{O}$ is "Sentence: $[x_j]$ ". Afterward, we calculate the cosine similarity between the embeddings of x_i and x_j . As a result, the sample most similar to x_i is decided as the reference sample x^* .

(3) **LLM choice-based selection:** This strategy leverages LLM APIs to choose the most similar sample. Due to the absence of labels for novel categories, LLMs cannot query directly to decide categories. Instead, we leverage a scalable query strategy [4] that asks the LLM to identify the true neighbors of $x_i \in \tilde{\mathcal{S}}_\delta$ from $\mathcal{N}(x_i)$. This strategy can be scalable to the number of candidate neighbors. An example prompt is: "Select the sentence that better corresponds with the query sentence. Query: $[S]$. Sentence 1: $[S1]$; Sentence 2: $[S2]$; ...; Sentence $|q|$: $[SN]$ ", where $[S]$ is the query sample x_i and $[S1] \dots [SN]$ are candidate neighbors. We denote the returned sample as the reference sample x^* .

Following studies [1, 13, 30], we optimize the representations of ambiguous samples, pulling them closer to their true neighbor samples. Then, the sample-sample contrastive learning is:

$$\mathcal{L}_{ss} = - \sum_{x_i \in \tilde{\mathcal{S}}_\delta} \log \frac{\exp(F_\theta(\mathcal{A}(x_i))^T F_\theta(\mathcal{A}(x^*)) / \tau_2)}{\sum_{x_j \in \mathcal{N}(x_i)} \exp(F_\theta(\mathcal{A}(x_i))^T F_\theta(\mathcal{A}(x_j)) / \tau_2)}, \quad (16)$$

where $\mathcal{A}(\cdot)$ is the data augmentation to improve generalizability, $\mathcal{N}(x_i)$ is the neighbors. τ_2 is the temperature factor.

4.4 Training Objective and Prediction

To retain the discriminating capability of known categories, we also add the cross-entropy loss $\mathcal{L}_c$ to the few available labeled data. Then, the overall objective $\mathcal{L}$ is combined with the disentanglement constraints during self-training iterations:

$$\mathcal{L} = \mathcal{L}_{ce} + \mathcal{L}_{ct} - \alpha \mathcal{L}_{em} + \beta \mathcal{L}_{cl}, \quad (17)$$

where $\mathcal{L}_{cl} = \mathcal{L}_{cs} + \mathcal{L}_{ss}$ is the contrastive loss in explicit disentanglement, and α and β are harmonious factors to balance losses. For prediction, we obtain representations of all samples, and perform K-mean clustering to assign categories.

Table 1: Statistics of the benchmark datasets.

Dataset	Domain	C	Samples
BANKING	BANKING	77	13,083
CLINC	Multi-Domain	150	22500
StackOverflow	Programming	20	22000

5 Experiment Setup

In this section, we conduct experiments to evaluate our proposed EMCD model with existing state-of-the-art methods.

5.1 Datasets

We evaluate our proposed model on three standard Generalized Category Discovery (GCD) benchmarks:

- **BANKING** [6] is a fine-grained intent dataset constructed from banking domain user dialogues, comprising over 13,000 utterances across 77 distinct categories.
- **CLINC** [12] is a multi-domain dataset spanning 10 domains, containing 22,500 utterances over 150 intent categories.
- **StackOverflow** [23] is a technical Q&A dataset sourced from Kaggle, including 20,000 questions categorized into 20 topics.

Following the established experimental protocol [15], we treat 25% or 50% of the categories as the known category rate (KCR), with 10% of the data serving as labeled samples. We use all data for testing. The detailed dataset statistics and splits are summarized in Table 1.

5.2 Baselines

We conduct a comprehensive comparison of our model against a range of state-of-the-art methods, which we group into two primary categories based on their core methodologies: **Contrastive Learning-based Models**:

- **DeepAligned** [26] addresses clustering inconsistency through a text embedding alignment strategy in semi-supervised learning.
- **MTP-CLNN** [29] employs a two-stage approach using multi-task pre-training and nearest-neighbor contrastive learning to identify new categories.
- **USNID** [27] utilizes a centroid initialization strategy leveraging historical clustering information to improve cluster representations in both semi-supervised and unsupervised settings.
- **CsePL** [13] introduces a two-level contrastive learning framework with label semantic alignment and soft prompting for novel intent discovery.
- **LOOP** [1] uses refined neighborhood contrastive learning to enhance sample proximity and clustering-friendly representations.
- **SynCID** [14] uses LLMs to refine utterances and intent labels, improving semantic depth and correcting cluster distortions in feature space.

Pseudo-labeling-based Models:

- **ProbNID** [31] proposes a probabilistic framework to assign intent pseudo-labels as latent variables.
- **ALUP** [15] integrates LLM feedback into active learning, with uncertainty propagation selecting high-uncertainty samples for comparison-based labeling.

All the experimental results in Table 2 for these baseline models are obtained from the benchmark dataset reported in Liang et al. [15]. For a fair comparison, we use the same base model and K-means clustering as the baselines.

5.3 Evaluation Metrics

We evaluate our model, EMCD using three standard metrics: Accuracy (ACC) via the Hungarian algorithm, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI). Detailed definitions are provided in Appendix A.1, with ACC serving as the primary metric where higher values indicate better performance.

5.4 Implementation Details

For the fair comparison, we use the pre-trained bert-base-uncased model [10], featuring a 12-layer transformer architecture, as the foundational SLM for training. The GPT3.5 Turbo API and Qwen3-Embedding are used as the LLM and random token replace is used for data augmentation, following previous works [1, 29]. The pre-training epoch is set to 100 and the training epoch is set to 50. The learning rate for pretraining and training is set to $5e-5$ and $1e-5$, respectively. For masked language modeling, the mask probability is set to 0.15. For hyper-parameters of the second training process, the threshold γ is tuned in $[0, 1]$. The temperature parameters τ_1 and τ_2 are tuned in $[0.01, 0.09]$. The harmonic factor α is tuned in $\{0.0001, 0.001, 0.01, 0.1, 1\}$ and β is tuned in $[0, 1]$. Note that the hyper-parameter settings are tuned on the test data by grid search with 3 trials. The detailed optimal hyper-parameters are provided in Appendix A.2 for re-implementation.

6 Overall Results

6.1 Main Results

Table 2 presents the principal results of our proposed model compared to the existing baseline methods on GCD. We evaluated the performance of our framework across multiple datasets, including the single-domain, fine-grained BANKING and Stack Overflow datasets, and the multi-domain CLINC dataset. The results demonstrate that our model achieves superior performance in the majority of datasets and evaluation metrics, confirming its overall effectiveness. Specifically, our approach consistently outperforms all existing baselines by a considerable margin in most scenarios. Comparing to NMI that quantifies clustering consistency, EMCD has a large improvement in NMI via EM-driven contrastive disentanglement. This indicates that our model exhibits enhanced capability in novel category discovery and effectively mitigates the bias towards known categories. We analyze the results as follows:

- *Comparison of different traditional methods*: EMCD (w/o LLMs) denotes the results of our model without incorporating large language modules, ensuring a fair comparison with traditional methods. Our model was evaluated against two major categories of representative mainstream methods. The results demonstrate

Table 2: Main performance results (%) on GCD across three public datasets: clinc, banking and stackoverflow. KCR denotes the known category rate. Results are averaged over three random runs. The state-of-the-art (SOTA) results of the existing methods are underlined. [†] denotes that the method uses LLM.

KCR	Method	CLINC			BANKING			StackOverflow			AVG
		ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	NMI	ACC
25%	DeepAligned	74.07	64.63	88.97	49.08	37.62	70.50	54.50	37.96	50.86	59.22
	MTP_CLNN	83.26	76.20	93.17	65.06	52.91	80.04	74.70	54.80	73.35	74.34
	ProbNID	71.56	63.25	89.21	55.75	44.25	74.37	54.10	38.10	53.70	60.47
	USNID	83.12	77.95	94.17	65.85	56.53	81.94	75.76	65.45	74.91	74.91
	CsePL	86.16	79.65	94.07	71.06	60.36	83.22	79.47	64.92	74.88	78.90
	Loop [†]	86.58	80.67	94.38	71.40	60.95	83.37	82.20	66.29	79.10	80.06
	ALUP [†]	<u>88.40</u>	<u>82.44</u>	94.84	74.61	62.64	84.06	82.20	64.54	76.58	81.73
	SynCID [†]	87.85	82.39	<u>94.85</u>	<u>75.41</u>	<u>65.40</u>	<u>85.39</u>	<u>87.86</u>	<u>76.11</u>	<u>82.46</u>	<u>83.71</u>
	Ours (w/o LLM)	86.84	81.00	94.43	72.24	60.49	83.84	86.70	72.94	81.98	81.93
	Ours [†]	88.42	82.49	94.99	75.83	64.74	86.32	88.11	76.76	83.64	84.12
50%	DeepAligned	80.70	72.56	91.59	59.38	47.95	76.67	74.52	57.62	68.28	71.53
	MTP_CLNN	86.18	80.17	94.30	70.97	60.17	83.42	80.36	62.24	76.66	79.17
	ProbNID	82.62	75.27	92.72	63.02	50.42	77.95	73.20	62.46	74.54	72.95
	USNID	87.22	82.87	95.45	73.27	63.77	85.05	82.06	71.63	78.77	80.85
	CsePL	88.66	83.14	95.09	76.94	66.66	85.65	85.68	71.99	80.28	83.76
	Loop [†]	<u>90.98</u>	85.15	95.59	75.06	65.70	85.43	85.90	72.45	80.56	83.98
	ALUP [†]	90.53	84.84	95.97	79.45	68.78	<u>86.79</u>	86.70	73.85	81.45	85.56
	SynCID [†]	90.64	<u>85.96</u>	95.91	77.83	67.67	86.33	<u>88.40</u>	<u>77.24</u>	<u>83.34</u>	85.62
	Ours (w/o LLM)	91.08	86.57	96.17	77.42	67.31	86.05	86.64	76.64	82.53	85.05
	Ours [†]	92.31	87.41	96.44	78.25	67.72	86.34	89.40	78.92	85.04	86.65

that our method surpasses all traditional baselines; particularly in the CLINC dataset, it outperforms the conventional CsePL method by a margin of 2.42% in accuracy, even exceeding the sota result, which leverages large language models.

- *Comparison of different methods with LLMs:* EMCD demonstrates significantly superior performance compared to its variant without large language models (EMCD w/o LLM). When utilizing the same base large language model, our approach outperforms other LLM-enhanced baseline methods, including Loop, ALUP, and SynCID. This indicates that our Expectation-Maximization (EM) disentanglement process effectively enhances the model’s capability in novel category discovery tasks. Notably, even without employing any large language model, our method surpasses the LLM-based approaches Loop and SynCID across all three datasets when the known category ratio (KCR) is set to 50%.

6.2 Ablation Study

To assess the contribution of each component, an ablation study was conducted on the CLINC dataset through the individual removal of constituent modules. The corresponding results, averaged over three independent runs on test dataset under the conditions of $KCR = 25\%$ and $KCR = 50\%$, are summarized in Table 3.

Since existing methods [2, 4, 15] have established that base model pre-training is fundamental to GCD, our experiments are confined to the ablation of subsequent modules. The architecture incorporates two primary modules: Latent Category Disentanglement module and Explicit Category Disentanglement module. Notably, the

Table 3: Ablation results on the CLINC dataset.

KCR	Ablation	CLINC		
		ACC	ARI	NMI
25%	EMCD	88.42	82.49	94.99
	w/o EM	80.04	73.90	92.66
	w/o sample-sample	84.93	78.99	93.92
	w/o cluster-sample	86.71	81.14	93.54
50%	EMCD	92.31	87.41	96.44
	w/o EM	90.49	86.68	96.32
	w/o sample-sample	89.42	85.63	96.04
	w/o cluster-sample	91.47	87.62	96.42

Explicit Category Disentanglement module is composed of two constituent components: the Cluster-Sample Contrastive Constraint module and the Sample-Sample Contrastive Constraint module. As evidenced by the results, the performance declined after the removal of each individual module, regardless of whether under the $KCR = 25\%$ or 50% setting. Notably, under the $KCR = 25\%$ condition, removing the Expectation-Maximization Driven Contrastive Disentanglement module, led to a performance decrease of 8.38%.

6.3 Effect of Different LLMs

To verify different selection strategies, we evaluate three strategic implementations within the sample-to-sample contrastive learning

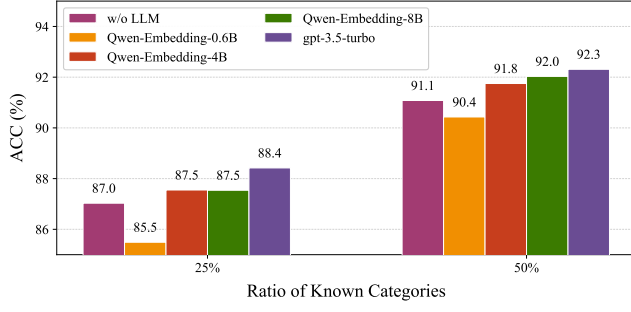


Figure 3: Results of different strategies to select refined samples on CLINC.

module on the CLINC benchmark: a baseline without large language model integration, an approach incorporating a lightweight clustering LLM, and a method utilizing a proprietary LLM. The superior performance of state-of-the-art GCD techniques stems from their use of LLMs to supply supervisory signals for novel categories, which are inherently unlabeled. Accordingly, we compare our framework against the dominant paradigm of API-based access to proprietary LLMs. To maintain comparability, the widely-used GPT-3.5 Turbo serves as the reference proprietary model in our experiments. For the lightweight LLM, we adopt the Qwen-Embedding model [24], fine-tuned specifically for our task.

As shown in Figure 3, the variant employing the proprietary LLM achieves the best performance under all KCR settings. Notably, even without any LLM, our baseline remains competitive with LLM-based variants. In particular, under $KCR = 50\%$, the gap between the lightweight Qwen-Embedding-8B and proprietary GPT-3.5 Turbo is only 0.3%, underscoring the practical efficiency of our approach.

Table 4: Error Rates across Different Data Types on CLINC.

KCR	Method	Known	Novel	K↔N
25%	Pretrain	16.32	35.12	26.50
	CsePL	7.42	15.98	10.54
	LOOP	6.36	15.69	9.63
	Ours	7.76	14.71	6.48
50%	Pretrain	13.24	26.67	19.91
	CsePL	6.76	16.71	8.53
	LOOP	3.91	15.56	8.59
	Ours	5.51	9.47	4.90

6.4 Error Analysis

In this section, we perform an error analysis to examine the bias toward known categories and to evaluate the effectiveness of our proposed method. As shown in Table 4, we present the ratio of novel samples that are misclassified as known samples by the model. In Table 4, "Known" denotes the error rate for known categories, "Novel" indicates the error rate for novel categories, and "K↔N" denotes the entanglement error between known and novel categories.

For comparison, we include state-of-the-art methods CsePL (without LLM) and LOOP on CLINC. Results indicate that the GCD task has novel error 35.12% and "known and novel" distanglement error 26.5% due to the lack of supervision for novel categories. Even LLM-based methods show no significant improvement in disentanglement errors between known and novel categories. In contrast, our approach achieves consistently lower error rates for novel categories and for disentanglement between known and novel categories, under both $KCR = 25\%$ and $KCR = 50\%$ settings.

6.5 Impact of Selected Samples Center Ratio γ

The hyper-parameter γ controls the representation sampling ratio, which will be used to compute category prototypes. Specifically, a higher γ encourages selecting more samples as center and a lower γ encourages less samples as center. γ is an essential hyper-parameter for our framework to guarantee the accurate categories prototypes. To examine its impact on performance, we conduct experiments with three independent runs under varying known category ratios (KCRs). As shown in Figure 4(a) and (b), results across different KCRs indicate that $\gamma = 0.1$ yields the best performance.

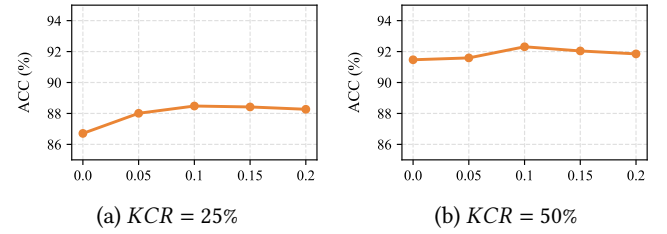


Figure 4: Results of different γ on CLINC.

7 Conclusion

In this paper, we present an EM driven contrastive disentanglement framework, EMCD, for the GCD task. Existing studies usually suffer from an inherent bias toward known categories, which can lead to significant misclassification and cluster errors for novel categories. To mitigate this limitation, we propose a latent category disentanglement module with EM-disentangling regularization, which explicitly models the distinction between novel and known categories. Besides, we leverage an explicit category disentanglement module, which pulls close samples within the cluster of novel categories, and pushes apart different clusters. Experimental results demonstrate that EMCD outperforms state-of-the-art models, and our model without LLM can even achieve competitive results with existing LLM-based methods. Our future work would further leverage LLMs to enhance the discrimination of novel category samples.

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A Appendix

In this appendix, we provide: (i) the evaluation protocol, and (ii) optimal hyper-parameters.

A.1 Evaluation Protocol

We evaluate the GCD performance using three standard metrics: Accuracy (ACC) with Hungarian algorithm mapping, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI) [15, 29]. Accuracy (ACC) measures the proportion of correctly classified samples after resolving label permutations via the Hungarian algorithm as follows:

$$ACC = \frac{\sum_{i=1}^N \mathbb{I}y_i = \text{map}(\hat{y}_i)}{N}, \quad (18)$$

where $y_i, \hat{y}_i$ denote the ground-truth and predicted labels for sample x_i , and $\text{map}(\cdot)$ aligns predicted labels to ground-truth labels using the Hungarian algorithm. Besides, ARI assesses pairwise clustering similarity by comparing agreements against chance, which is defined as:

$$ARI = \frac{\sum_{i,j} \binom{n_{ij}}{2} - \frac{\left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right]}{\binom{n}{2}}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \frac{\left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right]}{\binom{n}{2}}}, \quad (19)$$

where $u_i = \sum_j n_{i,j}$, and $v_j = \sum_i n_{i,j}$. N denotes the number of all samples. $n_{i,j}$ is the number of sample pairs that are both assigned to i th predicted cluster and j th ground-truth cluster. NMI quantifies clustering consistency via information theory:

$$NMI(\hat{y}, y) = \frac{2 \cdot I(\hat{y}, y)}{H(\hat{y}) + H(y)}, \quad (20)$$

where $I(\hat{y}, y)$ is the mutual information between $\hat{y}$ and y . $H(\cdot)$ represents the entropy function.

Hyper-parameters	CLINC	BANKING	StackOverflow
batch size	128	128	128
pretrain learning rate	$5e^{-5}$	$5e^{-5}$	$5e^{-5}$
train learning rate	$1e^{-5}$	$1e^{-5}$	$1e^{-5}$
pretrain epoch	100	100	100
epoch	50	50	50
embedding size d	256	256	256
number K	50	50	500
ratio γ	0.1	0.1	0.15
temperature τ_1	0.04	0.04	0.04
temperature step τ_2	0.07	0.07	0.07
factor α	0.01	0.0001	0.0001
factor β	1	0.5	0.2

Table 5: Detailed hyper-parameters of our EMCD model.

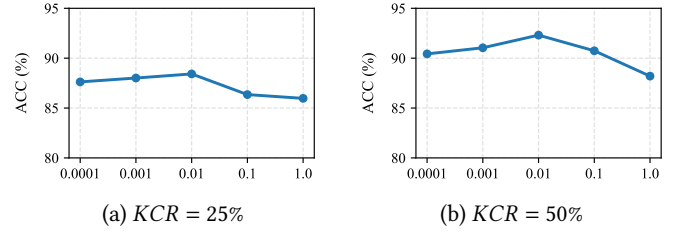
A.2 Optimal Hyper-parameters

We implement our model with Pytorch¹. Experiments are conducted on a server with NVIDIA V100 GPUs. The optimal model is selected according to the average ACC on all datasets by grid-search with three trials. We use Adam [11] to learn the model. The tuning ranges of hyper-parameters are reported in the main text. Here, we report the hyper-parameters used in Table 5.

A.3 Impact of Hyper-parameter Correlation Coefficient α

In our methodology, the hyper-parameter α is employed as a correlation coefficient to regulate the expectation-maximization (EM)

disentanglement ratio, which governs the balance between prior knowledge integration and data-driven optimization during the disentanglement process. To ensure statistical reliability, all experiments were conducted three times independently on the test set, with consistent outcomes observed across repetitions. As illustrated in Figure 5, the results demonstrate that an α value of 0.01 consistently achieves optimal performance under both scenarios where the known category proportion is set to 25% or 50%. This result indicates that by iteratively refining the estimation of latent variables with a minimal α , the EM procedure promotes alignment with the latent data structure, thereby alleviating biases towards known categories.

**Figure 5: Results of different α on CLINC.**

¹<https://docs.pytorch.org/docs/stable/index.html>